

Individual Assessment Coversheet

To be attached to the front of the assessment.

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Faculty: Information Technology

Module Code: ITNSA2-11

Group: 1

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Indicate	Yes	No
Plagiarism report attached		

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Section A

Question 1

1.1) Implementing the FTP service in a manner that exposes it to the internet, without protecting it with a security device like a firewall, increases the potential for BusyBugs to become a target to cyber attackers. Cyber attackers will take interest in the network as they will identify that the FTP service is a network vulnerability and can then create a plan to utilise this vulnerability to their advantage. Moreover, the lack of authentication by enabling anonymous access further increases the chances of the network becoming a target for cyber attacker, because attackers will already have access to the network without having to perform certain attacks.

The location of the FTP service further compromises the security of the network, due to it being located within the management subnet. After the cyber attackers used the anonymous access feature of the FTP service to gain access to the network, they could utilise the network management and administrative features of the management subnet to spread their attack across the network efficiently and effectively before being detected. In addition, the improper network segmentation also enabled the attackers to easily gain more access to the network as whole when compared to networks with proper segmentation procedures and measures. Therefore, a detailed plan that specifies what aspects of the network should be altered to minimise vulnerabilities and prevent future network attacks should be developed.

In this plan, the FTP service must be relocated in the network where it is not directly exposed to the internet. A firewall device should be implemented to allow remote users to access this service while the firewall protects against potential breaches by blocking all traffic that aren't necessary for the users to perform their work. The FTP service should also be configured to replace the anonymous access feature with authentication mechanisms that forces users to prove their identities to the network. Authentication will greatly improve the overall security of the network, because attackers will have to go through additional hurdles if they want to breach the network as users of BusyBugs.

Various access control measures can be implemented alongside the authentication to prevent or minimise the potential impact of successful breaches. Role-Based Access Control (RBAC) is a security measure that determines what network resources a user will have access to based on their role within the network. This measure ensures that specific resources can only be accessed by authorised users (Singh, J., Rani, S., & Kumar, V., 2024: 96). RBAC also know as authorization, can be beneficial for BusyBugs, because it can prevent attackers from gaining access to critical features and information within the network. This can lesson the potential impact of a successful breach due to the lack of permissions that a type of user that they're impersonating has. Moreover, Access Control Lists (ACL) can be created and configured for session filtering that prevents potential, outside malicious users from connecting to the network while not impacting devices already inside the protected network (Palugyai, S., 2005: 186). Additionally, a Virtual Private Network (VPN) can be implemented by BusyBugs to improve the security of the connection for remote users where the ACLs can be configured to only allow outside devices that uses the business's VPN.

Furthermore, network segmentation can be improved upon with new techniques that will also enhance the security of the network as a whole. A Virtual Local Area Network (VLAN) refers to a

technique where a network is segmented into smaller subnets, and it can be implemented for each subnet in the network to improve security. Security is improved by creating VLANs, because it enables network administrators to have enhanced control over each port on the physical devices, like switches, and users. Additionally, it also prevents on-site attackers from accessing the entire subnet through a singular port on the physical devices (Makeri, Y. A., Cirella, G. T., Galas, F. J., Jada, H. M., & Adeniran, A. O., 2021: 4755). In addition, microsegmentation techniques can also be utilised to protect against lateral movements in the network due to it enforcing security in each individual with a central policy. Microsegmentation techniques that are beneficial for BusyBugs include host-agent segmentation, where all data flows are visible to a central manager and discover suspicious traffic, as well as network segmentation, where segmentation is built from the network's ACLs (Edwards, J., 2020).

Moreover, monitoring solutions should also be implemented to detect and minimise the damage of successful breaches in the network. Intrusion detection technology should be used to monitor the network, because it's able to analyse the network for suspicious activity and traffic such as lateral movements. A type of intrusion detection system that BusyBugs can apply to their network is a host-based intrusion detection system, where this system will detect intrusions and attacks by creating and analysing logs of the internal operations of each computer system (Sun, Y., Liu, X. & Shen, X., 2022: 4).

1.2) A monitoring and incident response plan can be created and implemented to mitigate the damage of future network breaches. Various aspects should be taken into consideration when creating the monitoring and incident response plan such as conducting an incident threat analysis to determine potential vulnerabilities in the network that should be planned for. Moreover, this plan should be constantly reviewed and tested to determine areas that can be updated to improve its effectiveness when recovering from attacks (Drinkwater, D., 2017).

In addition, computers within the network should be configured to record the state of resources in real-time through logs. These logs should be viewable by system administrators that enables them to analyse it to determine if the computer system demonstrates abnormal activities (Sun, Y., Liu, X. & Shen, X., 2022: 4). Additionally, intrusion detection systems (IDS) and intrusion prevention systems (IPS) can be implemented to the network to prevent and detect attacks. An IPS is used to identify malicious traffic from attackers and block it before they are able to breach the network. An IDS is used to detect successful breaches in the network by notifying the business of suspicious activities, enabling them to take action perform the incident response plan.

A well-defined alerting and incident response process can assist in preventing future lateral movement and data exfiltration, because it provides the guidelines for the business to investigate how attackers were able to breach and move through the network while stealing data and implement new security mechanisms that minimises the vulnerabilities that were exploited. Furthermore, new vulnerabilities that are unrelated to the successful breach can be discovered and removed before future attackers can exploit it to gain access to the network.

Question 2

2.1) Additional security measures can be implemented into the SafeNet's VPN such as multi factor authentication (MFA). Implementing multiple authentication methods can greatly improve VPN security, because attackers will be less likely to obtain all the required authentication factors. In addition to the increase in difficulty for attackers to gain access to the network through brute force attacks, MFA also reduces the amount of potential false alarms that can cause disruptions (Ayman, M.M., Ezz, M., Elbashir, M.K., Alruily, M., Hamouda, E., Alsarhani, M. & Said, W., 2023: 7).

Moreover, an access control list can be applied and configured for the VPN and RDP service to block unwanted traffic from remote users to the network. The access control list can be configured to block specific traffic that is not required for the remote users to perform their tasks, which prevents attackers from utilising it to gain access or disrupt the system.

Furthermore, the corporate LAN can be further segmented by dividing the network into multiple zones to separate each of SafeNet's departments such as Accounting, IT and Customer Support. Separating each department through network segmentation enables SafeNet to contain any successful brute force attacks and prevent the attackers from further gaining access to additional areas in the network. Network segmentation should be done by implementing VLANs, because it allows SafeNet to implement segmentation with existing infrastructure and be configured to best match their requirements.

After segmenting the network, a specific firewall policy can be created for each segmented zone where each policy will have different rules depending on the zone. An area in the network that contains sensitive data should have an intrusive firewall policy with severe restrictions to protect the data from unauthorised access, however other network areas might require a less restrictive firewall policy to allow users and employees to work efficiently.

2.2) Various encryption protocols can be utilised by SafeNet to improve the security of their VPN and RDP service such as secure shell (SSH) and internet protocol security (IPsec). SSH can be used to improve VPN security while maintaining performance due to it automatically encrypting data transmitted between the remote users and the network. Encrypting the data protects the communication between users and the network from being eavesdropped and modified by attackers (Okoh, C., Waba, N.T., Dawkins, P. & Paheerathan, S., 2024: 29).

IPSec refers to a suite of protocols that provides security through IP extensions. IPSec supports authentication header (AH) and encapsulation security payload (ESP) where AH ensures data integrity and ESP provides confidentiality alongside the services offered by AH, which can be quite beneficial for the VPN and RDP service of the network. Moreover, IPSec can be implemented in transport mode where only the packet payload is protected, which offers protection while maintain efficiency (Thiruvassagam, P. & George, K.J., 2019: 61).

Additionally, firewalls can also be configured to secure SafeNet's network while maintaining performance by implementing the principle of least privilege. By configuring the firewall's settings to only allow users to only perform certain actions related to their position in SafeNet can greatly minimise the potential damage of a successful brute force attack despite the attackers appearing as authorised users. Moreover, a firewall can also block certain types of traffic that an authorised user wouldn't need, further minimising any potential damage.

2.3) The process of monitoring and logging instances of traffic within a network enables businesses like SafeNet to identify breaches through unauthorised activities. Traffic monitoring and logging can be beneficial for the security of a network as it enables businesses to discover undetected security breaches and minimise the potential damage that could have been done to the network if it was never identified.

An intrusion detection system (IDS) can be implemented by SafeNet to monitor the traffic in their hybrid network and notify them of any traffic that it detects as suspicious. Additionally, an intrusion prevention system (IPS) acts similarly to an IDS system where it will automatically block suspicious traffic that it detects, allowing for SafeNet to investigate it afterwards. An IPS system is the most suitable for SafeNet's hybrid network infrastructure, because the system analyses traffic outside of the network that enables it to identify and block brute force attacks before it can breach the network. An IDS system can only analyse traffic in the network, allowing successful brute force attacks to breach the network and do damage before it can be detected.

Question 3

3.1) The 10.20.0.0/24 network should be subdivided into four distinct VLANs where each department will be assigned to a single VLAN. These VLANs will enable each device in each department to remain logically isolated from each other with requiring additional physical components, which improves the scalability for Karoo Manufacturing's network.

Administration subnet:

- Vlan 20
- Subnet: 10.20.20.0/24
- IP range: 10.20.20.2 – 10.20.20.255
- Subnet mask: 255.255.255.0

Sales subnet:

- Vlan 30
- Subnet: 10.20.30.0/24
- IP range: 10.20.30.2 – 10.20.30.255
- Subnet mask: 255.255.255.0

Production subnet:

- Vlan 40
- Subnet: 10.20.40.0/24
- IP range: 10.20.40.2 – 10.20.40.255
- Subnet mask: 255.255.255.0

HR subnet:

- Vlan 50
- Subnet: 10.20.50.0/24
- IP range: 10.20.50.2 – 10.20.50.255
- Subnet mask: 255.255.255.0

The VLANs are created in the main switch of the internal network with the following commands:

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/
Switch(config)#vlan 20
Switch(config-vlan)#name Administration
Switch(config-vlan)#exit
Switch(config)#vlan 30
Switch(config-vlan)#name Sales
Switch(config-vlan)#exit
Switch(config)#vlan 40
Switch(config-vlan)#name Production
Switch(config-vlan)#exit
Switch(config)#vlan 50
Switch(config-vlan)#name HR
Switch(config-vlan)#exit
```

```

Switch(config)#interface fastethernet0/1
Switch(config-if)#switchport access vlan 20
Switch(config-if)#switchport mode access
Switch(config-if)#exit
Switch(config)#interface fastethernet0/2
Switch(config-if)#switchport access vlan 30
Switch(config-if)#switchport mode access
Switch(config-if)#exit
Switch(config)#interface fastethernet0/3
Switch(config-if)#switchport access vlan 40
Switch(config-if)#switchport mode access
Switch(config-if)#exit
Switch(config)#interface fastethernet0/4
Switch(config-if)#switchport access vlan 50
Switch(config-if)#switchport mode access
Switch(config-if)#exit

```

Each switch within each VLAN should also be configured to access the corresponding VLAN with the above commands.

The main switch of the internal network should be connected to the router with the following configuration:

```

Switch(config)#interface GigabitEthernet0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit

```

To enable the devices in each department to communicate with each other, the router for the internal network must be configured to have IP addresses for each VLAN. It will allow the router to route traffic between the VLANs with the following commands:

```

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitethernet0/1
Router(config-if)#ip address 10.20.0.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit

Router(config)#interface gigabitethernet0/1.20
Router(config-subif)#encapsulation dot1q 20
Router(config-subif)#ip address 10.20.20.1 255.255.255.0
Router(config-subif)#exit

Router(config)#interface gigabitethernet0/1.30
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1.30, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1.30, changed state to
up

Router(config-subif)#encapsulation dot1q 30
Router(config-subif)#ip address 10.20.30.1 255.255.255.0
Router(config-subif)#exit

Router(config)#interface gigabitethernet0/1.40
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1.40, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1.40, changed state to
up

Router(config-subif)#encapsulation dot1q 40
Router(config-subif)#ip address 10.20.40.1 255.255.255.0
Router(config-subif)#exit

```



```

Router(config)#interface gigabitethernet0/1.50
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1.50, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1.50, changed state to
up

Router(config-subif)#encapsulation dot1q 50
Router(config-subif)#ip address 10.20.50.1 255.255.255.0
Router(config-subif)#exit

```

3.2) The web and email server must be assigned their respective IP addresses as well as the IP address of the router as the default gateway.

Web Server

Physical Config Services **Desktop** Programming Attributes

IP Configuration

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 10.20.10.10

Subnet Mask: 255.255.255.0

Default Gateway: 10.20.10.1

DNS Server: 0.0.0.0

Email Server

Physical Config Services **Desktop** Programming Attributes

IP Configuration

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 10.20.10.20

Subnet Mask: 255.255.255.0

Default Gateway: 10.20.10.1

DNS Server: 0.0.0.0

The inside and the outside of the whole network should be specified in the firewall to determine the security level of each aspect of the network. The internal network should be specified as inside the network with a security level of 100, that indicates that it's trusted by the network, while the external network is considered outside the network and has a security level of 0, that indicates that it's not trusted by the network. This is done by the following commands:

```

ciscoasa(config)#interface gigabitethernet1/1
ciscoasa(config-if)#nameif inside
ciscoasa(config-if)#security-level 100
ciscoasa(config-if)#exit
ciscoasa(config)#interface gigabitethernet1/3
ciscoasa(config-if)#nameif outside
ciscoasa(config-if)#security-level 0
ciscoasa(config-if)#exit

```

NAT can then be implemented after establishing the route between the DMZ, external and internal network. It should also lead to the internet through the IP address of the external network port in the firewall with the following commands.

```
ciscoasa(config)#route inside 10.20.0.1 255.255.255.0 10.20.90.2
ciscoasa(config)#route outside 0.0.0.0 0.0.0.0 10.20.255.2
```

NAT will then be used to protect the IP addresses of the devices and servers within the network by using the same public IP address with the following commands:

```
ciscoasa(config)#object network INSIDE-NET
ciscoasa(config-network-object)#subnet 0.0.0.0 0.0.0.0
ciscoasa(config-network-object)#nat (inside, outside) dynamic interface
ciscoasa(config-network-object)#exit
```

The ACLs should then be created, where all traffic is denied except for traffic that is specified in the ACL depending on the device. The ACLs should permit HTTP (80) and HTTPS (443) traffic for the web server, permit SMTP (25) and POP3 (110) for the email server and deny traffic from the DMZ and external network that is forwarded to the internal network with the following commands.

```
ciscoasa(config)#access-list 101 deny tcp any any
ciscoasa(config)#access-list 101 permit tcp host 10.20.10.10 host 10.20.255.1 eq 80
ciscoasa(config)#access-list 101 permit tcp host 10.20.10.10 host 10.20.255.1 eq 443
ciscoasa(config)#access-list 101 permit tcp host 10.20.10.20 host 10.20.255.1 eq 25
ciscoasa(config)#access-list 101 permit tcp host 10.20.10.20 host 10.20.255.1 eq 110
ciscoasa(config)#access-list 101 deny tcp 10.20.0.0 255.255.255.0 host 10.20.255.1
ciscoasa(config)#access-list 101 deny tcp 10.20.0.0 255.255.255.0 host 10.20.80.1
```

(Greg South, 2020)

Question 4

4.2) Before the IPSec can be implemented on both routers, each router must have the security license enabled.

```
Router(config)#license boot module c1900 technology-package securityk9
Router#copy run start
Destination filename [startup-config]?
Building configuration...
[OK]
Router#reload
```

ACLs should be configured for the routers at both the Harrismith and Bloemfontein facility where the permit each other's networks (192.168.10.0/24 and 192.168.50.0/24).

Harrismith router:

```
Router(config)#access-list 101 permit ip 192.168.10.0 0.255.255.255 192.168.50.0 0.255.255.255
```

Bloemfontein router:

```
Router(config)#access-list 101 permit ip 192.168.50.0 0.255.255.255 192.168.10.0 0.255.255.255
```

The IKE phase 1 settings are then implemented on each router through ISAKMP policy.

Harrismith router:

```
Router(config)#crypto isakmp policy 10
Router(config-isakmp)#encryption aes 256
Router(config-isakmp)#authentication pre-share
Router(config-isakmp)#group 5
Router(config-isakmp)#exit
Router(config)#crypto isakmp key secretkey address 192.168.5.1
```

Bloemfontein router:

```
Router(config)#crypto isakmp policy 10
Router(config-isakmp)#encryption aes 256
Router(config-isakmp)#authentication pre-share
Router(config-isakmp)#group 5
Router(config-isakmp)#exit
Router(config)#crypto isakmp key secretkey address 192.168.1.1
```

The IPSec phase 2 settings are then implemented on each router.

Harrismith router:

```
Router(config)#crypto isakmp key secretkey address 192.168.5.1
Router(config)#crypto ipsec transform-set Harrismith-Bloemfontein esp-aes 256 esp-sha-hmac
Router(config)#crypto map IPSEC-MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
Router(config-crypto-map)#set peer 192.168.5.1
Router(config-crypto-map)#set pfs group5
Router(config-crypto-map)#set security-association lifetime seconds 86400
Router(config-crypto-map)#set transform-set Harrismith-Bloemfontein
Router(config-crypto-map)#match address 101
```


Bloemfontein router:

```
Router(config)#crypto isakmp key secretkey address 192.168.1.1
Router(config)#crypto ipsec transform-set Bloemfontein-Harrismith esp-aes 256 esp-sha-hmac
Router(config)#crypto map IPSEC-MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
Router(config-crypto-map)#set peer 192.168.1.1
Router(config-crypto-map)#set pfs group5
Router(config-crypto-map)#set security-association lifetime seconds 86400
Router(config-crypto-map)#set transform-set Bloemfontein-Harrismith
Router(config-crypto-map)#match address 101
```

A crypto map must then be assigned for each router.

Harrismith router:

```
Router(config-crypto-map)#interface gigabitethernet0/1
Router(config-if)#crypto map IPSEC-MAP
*Jan  3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
```

Bloemfontein router:

```
Router(config-crypto-map)#interface gigabitethernet0/1
Router(config-if)#crypto map IPSEC-MAP
*Jan  3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
```

(danscourses, 2018)

4.3) The connectivity between the Harrismith and Bloemfontein network should be tested to ensure that both networks are able to connect to each other through the VPN. This can be tested by pinging the PCs between each network.

PC in Harrismith network:

```
C:\>ping 192.168.50.2

Pinging 192.168.50.2 with 32 bytes of data:

Reply from 192.168.50.2: bytes=32 time<1ms TTL=126
Reply from 192.168.50.2: bytes=32 time<1ms TTL=126
Reply from 192.168.50.2: bytes=32 time<1ms TTL=126
Reply from 192.168.50.2: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.50.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

PC in Bloemfontein network:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time<1ms TTL=126
Reply from 192.168.10.2: bytes=32 time<1ms TTL=126
Reply from 192.168.10.2: bytes=32 time<1ms TTL=126
Reply from 192.168.10.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Furthermore, utilising the command “show crypto ipsec sa” in each of the routers enables us to determine if the VPN is properly encrypting and decrypting the packets that are sent between the networks.

PC in Harrismith network:

```
Router#show crypto ipsec sa

interface: GigabitEthernet0/1
  Crypto map tag: IPSEC-MAP, local addr 192.168.1.1

protected vrf: (none)
local  ident (addr/mask/prot/port): (192.0.0.0/255.0.0.0/0/0)
remote ident (addr/mask/prot/port): (192.0.0.0/255.0.0.0/0/0)
current_peer 192.168.5.1 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 15, #pkts encrypt: 15, #pkts digest: 0
#pkts decaps: 14, #pkts decrypt: 14, #pkts verify: 0
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 1, #recv errors 0
```

PC in Bloemfontein network:

```
Router#show crypto ipsec sa

interface: GigabitEthernet0/1
  Crypto map tag: IPSEC-MAP, local addr 192.168.5.1

protected vrf: (none)
local  ident (addr/mask/prot/port): (192.0.0.0/255.0.0.0/0/0)
remote ident (addr/mask/prot/port): (192.0.0.0/255.0.0.0/0/0)
current_peer 192.168.1.1 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 14, #pkts encrypt: 14, #pkts digest: 0
#pkts decaps: 15, #pkts decrypt: 15, #pkts verify: 0
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
```

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